Purab Balani

Email: balanipb@gmail.com | Phone: (949) 627-6457
LinkedIn: https://www.linkedin.com/in/purab-balani | Website: https://purabb.github.io

EDUCATION

University of California, San Diego: BS in Computer Engineering (Expected: December 2025 | GPA:3.63)
Coursework: Parallel Computing, Embedded Systems, Software Eng, Autonomous Vehicles, Intelligent Systems

ENGINEERING EXPERIENCE

Amazon, SDE Intern **June 2025 - September 2025**
Technologies Used: Java, gRPC, Protocolbuffer, aws EC2 instances, UDP, aws control plane/data plane architecture

- Accelerated propagation of a *1.5MB* health config file from a server node to *100K+ Top Client nodes*—from *3 minutes* to under *1 second*—by implementing **gRPC streaming.**
- Designed and deployed a custom load-balancing hash function to evenly distribute *100K+ clients across 5 servers*, with built-in failover protection and server-side rate limiting to prevent overloads.
- Conducted extensive load testing across multiple strategies (streaming vs polling, **gRPC vs UDP vs Nsync**) to identify the most efficient method for broadcasting data to 40K+ clients.
- Tuned **JVM heap sizes** and optimized **thread pool** configurations to reduce latency and improve throughput under high concurrency.
- Refactored legacy code to eliminate circular dependencies, ensuring reliable data propagation even during partial system failures.

Undergrad Research, Collaborative Intelligence Systems Lab (CISL) **January 2024 - December 2025**
Technologies Used: CNN, Lidar, Python, Linux, Pointpillar, Tensorflow, Cuda, VScode, Docker, Jetson Nano, OpenCV

- Developing a fully autonomous vehicle by integrating *vehicular communication, lidar, and camera sensing* to improve real-time perception and decision-making for autonomous driving.
  Generated **3D point-cloud** elements and incorporated them with **pointpillar** for object detection and used the data to train the ***Convolutional Neural Network*** **(CNN)** model for autonomous driving in **Python**.
- Utilizing visualization software to run simulations for autonomous driving and test and improve the model.
- Built an autonomous RC car for f1tenth competitions by adding lidar and cameras and training data on **Jetson Nano**

Undergrad Research, Robotics and Medical Systems (RaMS) **October 2022 - December 2023**
Technologies Used: Arduino, Matlab, Solidworks, Simulink, Altium, LTSpice, C++

- Designed a model and test bench of a *steerable microcatheter* and *brain surgery robot* using **Solidworks**.
- Implemented **Arduino** and **Simulink** to communicate inputs from sensors such as bend sensors and optic fibers to receive data from the medical device with **C/C++** to implement **PID (proportional-integral-derivative)**.
- Analyzed data to create a PID loop using **MATLAB** to control the motor *increasing accuracy to the nearest 0.05mm.*

PROJECTS

Autonomous GoKart Project **November 2023 - Current**
Technologies Used: CNN, Python, Linux, Camera, Altium, LTSpice, Solidworks, DonkeyCar, OpenCV, Jetson, ROS2

- Simulated and tested autonomous driving behaviors on an RC car using the **DonkeyCar** framework.
- Leveraged an **Nvidia Jetson** running **ROS2** to stream, synchronize, and process data from multiple cameras for autonomous-driving perception.
- Created PCB designs using **Altium** for power distribution to direct power to the 1800W drive motor alongside the onboard computer and other necessary components.

Autonomous Tank **Github: UCSD-ECEMAE-148/fall-2024-final-project-team-3: fall-2024-final-project-team-03**
Technologies Used: Python, Docker, Jetson Nano, Solidworks, DonkeyCar, Simulation, GNSS, OpenCV, Supercomputers

- Designed an autonomous tank using **GNSS** to drive on a set path, achieving a target detection accuracy of *95%*.
- Used **DonkeyCar** to create a dataset for the routes and ran simulations to set **PID** achieving a path accuracy of *90%*
- Utilized cameras and servos along with **OpenCV** to detect targets and point the laser based on its position.

Robotic Arm **Website:https://sites.google.com/ucsd.edu/lending-a-hand/home?authuser=1**
Technologies Used: ESP32, ESPNOW, Bluetooth, WiFi, PCB design, Altium, C++, Python, Solidworks, Arduino IDE

- Designed a tendon-driven arm capable of lifting weights around *3 kg* with an accuracy of *5 cm* by utilizing gear ratios.
- Implemented **Bluetooth** control using the *2.4 GHz* band to connect with an XBOX controller to control the arm.
- Used WiFi for ESPNOW to have multiple **ESP32s** send and receive packets, allowing for better rotation of freedom

Custom BLE Tracking Device **January 2025 - March 2025**
Technologies Used: STM32L4, BlueNRG-MS, BLE (HCI/ACI), C, I²C, LSM6DSL IMU, Low-Power Timers, Interrupts

- Built an ultra-low-power BLE tracker using an STM32L4, BlueNRG-MS radio, and LSM6DSL IMU, implementing interrupt-driven motion detection and custom HCI/ACI commands.
- Reduced power consumption from **~950 mA to 325 mA** using deep-sleep modes, low-power timers, and optimized BLE advertising

IBM AI Professional Certification **August 2024**